

Code-Layout, Readability and Reusability

Date	1 October 2026
Team ID	SWTID-2026-3138
Project Name	ComicCraft - AI Comic Story Creator using Gemini Models
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	4-space Python (PEP 8) indentation in all modules.
2	Proper File Structure	Files and folders are logically organized	Yes	app/ (logic), templates/ (HTML), static/panels and static/exports (output).
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Names like full_story, image_paths, pdf_path, story_panels.
4	Function / Method Names	Functions are descriptively named	Yes	generate_outline, generate_story, generate_image, build_comic_layout, save_pdf.
5	Code Comments	Inline and block comments explain logic	Yes	Docstrings on functions plus step-by-step comments in routes.py.
6	Modular Design	Code is split into reusable functions/modules	Yes	One module per task (outline, story, image, layout, export, routes).
7	No Redundant Code	Duplicate or unused code is removed	Partial	/generate and /generate-comic/json repeat the same pipeline; can move it to one shared function.
8	Error Handling	Exceptions and errors are handled gracefully	Yes	try-except, JSON decode checks, structure validation and HTTPException.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	gemini_flash.py - generate_outline()	Python, Google Generative AI	/generate and /generate-comic/json	High
2	gemini_pro.py - generate_story()	Python, Google Generative AI	/generate and /generate-comic/json	High
3	image_generator.py - generate_image()	Python, Diffusers, PyTorch	Both comic routes and /test-image	High
4	layout_builder.py - build_comic_layout()	Python	Both comic routes (preview and PDF)	Medium
5	exporters.py - save_pdf()	Python, FPDF	Both comic routes; any layout list	Medium

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	4	Clear folder structure and one module per task.
Readability	4	Descriptive names and consistent style.
Reusability	4	Functions are independent; the shared pipeline can still be merged.
Documentation / Comments	4	Docstrings and inline comments present; a README would help.
Overall Score	4 / 5	Well-organized, modular and easy to extend.